Supplementary Material: The Universal Explanation Impossibility

Drake Caraker Bryan Arnold David Rhoads

This supplement accompanies "The Universal Explanation Impossibility: No Explanation Is Faithful, Stable, and Decisive Under Underspecification." It contains: (A) full related work, (B) detailed proofs and experimental methodology for all six instances, (C) the extended Lean axiom inventory, and (D) defenses against common objections.

1 Full Related Work

Impossibility results in XAI. Bilodeau et al. [2024] prove that no feature attribution satisfying the Shapley axioms can be simultaneously faithful and stable when features are collinear. Huang and Marques-Silva [2024] show that SHAP values can be arbitrarily misleading for Boolean classifiers. Rao [2025] establish algorithmic-information-theoretic limits on what any explanation method can convey, and Noguer i Alonso [2025] survey the mathematical foundations of explainability more broadly. Each of these results is proved in a domain-specific language under domain-specific assumptions. Our contribution is a single abstract theorem that subsumes them as instances, showing the impossibility arises from underspecification itself rather than from any particular axiom system.

Underspecification and model multiplicity. D'Amour et al. [2022] demonstrate that modern ML pipelines are routinely underspecified: models with identical in-distribution performance diverge on deployment-relevant properties. Fisher et al. [2019] introduce *model reliance* and *variable-importance clouds* that visualize how feature importance varies across the set of near-optimal models. Rudin et al. [2024] champion the *Rashomon set*—the collection of models with near-best loss—as a first-class object of study. Semenova et al. [2022] prove the existence of simpler models within Rashomon sets, and Marx et al. [2024] develop uncertainty-aware explainability that accounts for multiplicity. Our framework formalizes the bridge between underspecification and explanation impossibility: the Rashomon property is precisely the condition under which equivalent hypotheses yield incompatible explanations.

Fairness impossibilities. Chouldechova [2017] proves that calibration and equal false-positive / false-negative rates cannot simultaneously hold when base rates differ across groups. Kleinberg et al. [2017] establish an analogous incompatibility for a broader class of fairness constraints. We recast these as an instance of the universal impossibility: the explanation system maps each classifier to a fairness certificate, and the Rashomon property holds whenever the underlying population has unequal base rates.

Formal verification of impossibility theorems. Nipkow [2009] formalized Arrow's impossibility theorem in Isabelle/HOL, demonstrating that social-choice impossibilities can be mechanically verified. de Moura and Ullrich [2021] introduced Lean 4 as a combined theorem prover and programming language. Zhang et al. [2026] recently formalized statistical learning theory in Lean 4. Our work provides, to our knowledge, the first Lean formalization of an impossibility theorem in explainability, covering all six instances and the G-invariant resolution (67 files, 319 theorems, 0 `sorry`).

Aggregation and resolution strategies. Laberge et al. [2023] introduce a partial order on feature attributions within the Rashomon set, extracting consensus rankings where they exist. Decker

Preprint.

et al. [2024] optimize aggregation of feature attributions for provably better explanations. Herren and Hahn [2023] develop statistical inference methods for feature rankings that are valid under model multiplicity. We show that these diverse strategies are all instances of the G-invariant resolution: they aggregate over the symmetry group of equivalent hypotheses, restoring stability and faithfulness-in-expectation at the cost of decisiveness.

2 Instance 1: Feature Attribution — Full Details

2.1 Explanation System

Fix $P \geq 2$ features partitioned into L collinearity groups. Within each group $\ell \in [L]$, every pair shares pairwise correlation $\rho \in (0, 1)$; features across groups are independent. The explanation system is:

- $\mathbf{\Theta}$: joint data-generating distributions with the collinearity structure above.
- $\mathcal{H}$: trained models from a fixed class (e.g., GBDT, Lasso, neural networks), indexed by random seed.
- $Y = (\mathrm{Fin}\, P \to \mathbb{R})$: attribution vectors $\boldsymbol{\varphi}$.
- $\mathsf{obs}(f) = \boldsymbol{\varphi}(f)$: the model's own attributions.
- exp: any attribution procedure.
- $\perp$: $\boldsymbol{\varphi} \perp \boldsymbol{\varphi}'$ iff $\exists\, j, k$ in the same group with $\varphi_j > \varphi_k$ and $\varphi'_k > \varphi'_j$.

2.2 Rashomon Property and Proof

Collinear features induce a Rashomon set of near-optimal models differing only in which feature within a correlated group acts as primary predictor. DGP symmetry within a group guarantees every feature can serve as first-mover in sequential optimization (surjectivity axiom).

Quantitative bounds. For GBDT with per-group correlation ρ, the attribution ratio diverges as $1/(1-\rho^2)$ (`Ratio.lean`). For Lasso at any $\rho > 0$, one correlated feature is zeroed entirely (ratio $= \infty$; `Lasso.lean`). For random forests with T trees, ensemble averaging yields $O(1/\sqrt{T})$ convergence—the mechanism behind DASH.

2.3 Resolution

DASH averages attribution vectors over an ensemble of retrained models, achieving consensus equity $\mathbb{E}[\varphi_j] = \mathbb{E}[\varphi_k]$ for symmetric features j, k. This is stable by construction and faithful in expectation; the cost is indecisiveness (collinear pairs are tied).

3 Instance 2: Attention Maps — Full Details

3.1 Explanation System

Let T be the number of tokens. The 6-tuple:

- $\mathbf{\Theta}$: weight configurations θ (Q, K, V, output projections).
- $\mathcal{H} = \Delta^T$: attention distributions.
- Y: input-output functions f_θ.
- $\mathsf{obs}(\theta) = f_\theta$; $\mathsf{exp}(\theta) = \alpha_\theta$.
- $\perp$: $\mathrm{argmax}_t\, \alpha_t \neq \mathrm{argmax}_t\, \alpha'_t$.

3.2 Rashomon Property

D'Amour et al. [2022] establish that networks trained to the same loss differ in internal representations. Jain and Wallace [2019] show attention weights can be dramatically permuted while leaving predictions unchanged.

3.3 Proof

By the Rashomon property, $\exists\, \theta_1, \theta_2$ with $f_{\theta_1} \approx f_{\theta_2}$ yet $\operatorname{argmax} \alpha_{\theta_1} \neq \operatorname{argmax} \alpha_{\theta_2}$. Theorem 1 of the main paper yields the impossibility. Lean: `attention_impossibility` in `AttentionInstance.lean`.

3.4 Experimental Methodology

Setup. 10 DistilBERT models with weight perturbations ($\sigma \in \{0.01, 0.02\}$) on 200 SST-2 sentences.

Results.

- Prediction agreement: 100% (all models classify every sentence identically).
- Argmax flip rate: 60.0% [95% CI: 59–61%].
- Mean Kendall τ: 0.30 [CI: 0.29–0.31].

Implications. Any attention visualization committing to a single most-attended token must either be unfaithful to some equivalent models or change when the model is retrained. The resolution: averaged attention rollout across the Rashomon set, sacrificing decisiveness.

4 Instance 3: Counterfactual Explanations — Full Details

4.1 Explanation System

- $\mathbf{\Theta}$: model parameters defining decision boundaries ∂f_θ.
- $\mathcal{H} = \mathcal{X}$: candidate counterfactuals for query x_0.
- $Y = \{0, 1\}^n$: test-set predictions.
- $\mathsf{obs}(\theta) = f_\theta|_{\text{test}}$; $\mathsf{exp}(\theta) = \arg\min_{x' : f_\theta(x') \neq f_\theta(x_0)} \|x' - x_0\|$.
- $\perp$: $(x_1' - x_0)^\top (x_2' - x_0) < 0$ (opposite directions).

4.2 Rashomon Property and Proof

Within $\mathcal{R}_\varepsilon$, models place boundaries on opposite sides of x_0, producing counterfactuals in opposite directions. Both achieve identical test predictions. Theorem 1 applies. Lean: `counterfactual_impossibility` in `CounterfactualInstance.lean`.

4.3 Experimental Methodology

Setup. 20 XGBoost models on German Credit (differing only in random seed). Mean AUC: 0.797, spread < 0.03. 45 query points denied credit.

Results.

- Direction flip rate: 23.5% [CI: 20.6–26.6%].
- Cross-model recourse validity: 61.3% [CI: 55.8–66.2%].
- Distance instability (CV): 0.328 [CI: 0.281–0.372].
- Most unstable feature: `checking_status` (43% flip rate).

Implications. Algorithmic recourse is fundamentally limited under model multiplicity. An individual may receive incompatible advice across equivalent models. The resolution: report Rashomon-robust recourses valid across $\mathcal{R}_\varepsilon$.

5 Instance 4: Concept Probes — Full Details

5.1 Explanation System

- $\mathbf{\Theta} = \Theta$: weight configurations.

- $\mathcal{H} = \mathbb{R}^c$: concept activation vectors (unit directions in c-dimensional representation space).
- Y: input-output functions f_θ.
- $\mathsf{obs}(\theta) = f_\theta$; $\mathsf{exp}(\theta) = v_\theta$ (CAV from linear probe).
- $\perp$: directions not related by rotation within the concept subspace.

5.2 Rashomon Property

Bolukbasi et al. [2016]: gender directions vary across architectures trained on the same corpus. Kim et al. [2018]: TCAV directions shift with random seed. Ravfogel et al. [2020]: iterative nullspace projection reveals multiple incompatible concept directions.

The zero-loss manifold has dimension $\geq d - n > 0$ for hidden dimension $d > n$, so infinitely many weight configurations produce identical input-output behavior but geometrically distinct representations.

5.3 Proof

Let θ_1, θ_2 witness the Rashomon property. Faithfulness: $\mathsf{exp}(\theta_i) = v_{\theta_i}$. Stability: $v_{\theta_1} = v_{\theta_2}$. But $(v_{\theta_1}, v_{\theta_2}) \in \perp$ implies $v_{\theta_1} \neq v_{\theta_2}$—contradiction. Lean: `concept_impossibility` in `ConceptInstance.lean`.

5.4 Experimental Methodology

Setup. 15 MLP classifiers (128–64 hidden, ReLU) on `sklearn load_digits` (8×8 images, 10 classes, 1797 samples), different random seeds. All achieve $> 95\%$ test accuracy.

Results.

- Prediction agreement: 97.2% [CI: 95.7–98.5%].
- Mean $|\cos(v_i, v_j)|$ (off-diagonal): 0.10.
- Concept direction instability ($1 - |\cos|$): 0.90 [CI: 0.89–0.91].
- TCAV score std (curved concept): 0.16; range 0.10–0.69 across seeds.

Implications. Concept directions are not model-invariant facts. Practitioners should report concept directions together with their variability across the Rashomon set.

6 Instance 5: Causal Discovery — Full Details

6.1 Explanation System

- $\mathbf{\Theta}$: DAGs over vertices V (data-generating structures).
- $\mathcal{H}$: DAGs over V (fitted explanations).
- Y: conditional independence structures (Markov equivalence classes).
- $\mathsf{obs}(h)$: CI structure (d-separation statements of DAG h).
- exp: any orientation rule returning a fully oriented DAG.
- $\perp$: opposite edge orientations ($X \to Y$ vs. $Y \to X$).

6.2 Markov Equivalence as the Rashomon Property

Verma and Pearl [1991]: two DAGs are Markov equivalent iff they share the same skeleton and v-structures. Any undirected edge in the CPDAG has DAGs oriented both ways in the equivalence class.

6.3 Proof

For any CPDAG with undirected edge $\{X, Y\}$, let h orient it as $X \to Y$ and h' as $Y \to X$. By Markov equivalence, $\mathsf{obs}(h) = \mathsf{obs}(h')$. By definition, $(h, h') \in \perp$. Theorem 1 applies. Lean: `causal_instance_impossibility` in `CausalInstance.lean` (3 axioms).

6.4 Resolution

Report the CPDAG: stable by construction, faithful in the sense that it captures all shared causal structure, but not decisive (undirected edges are unresolvable). This is formalized as `cpdag_is_stable` in `CausalDiscovery.lean`.

7 Instance 6: Model Selection — Full Details

7.1 Explanation System

- $\mathbf{\Theta}$: model configurations (hyperparameters, seeds, architectures).
- $\mathcal{H}$: trained models.
- Y: performance metrics with strict total order.
- $\mathsf{obs}(h)$: observed metric on evaluation set.
- exp: selection rule.
- $\perp$: opposite selection decisions.

7.2 Rashomon Property and Proof

The ε-Rashomon set $\mathcal{R}_\varepsilon = \{h : \mathsf{obs}(h) \geq \mathsf{obs}(h^*) - \varepsilon\}$ generically contains models with incompatible rankings on different evaluation samples. Theorem 1 applies. Lean: `model_selection_instance_impossibility` in `ModelSelectionInstance.lean`.

7.3 Experimental Methodology

Setup. 50 XGBoost classifiers (differing only in random seed) on German Credit (credit-g). Single fixed 80/20 training split; 20 independent random 80/20 evaluation splits.

Results.

- Mean AUC: 0.948 ± 0.0023.
- Unique "best" models: 11 / 50.
- Best-model flip rate: 80%.
- Consecutive flip rate: 89%.
- Mean AUC spread per split: 0.0197.

7.4 Design Space

Family A (decisive, unfaithful): standard argmax validation accuracy—recommends the worse-performing model on some splits. **Family B** (faithful, stable, indecisive): report the Rashomon set or ensemble predictions, using partial orders that render near-optimal models incomparable.

8 Extended Lean Axiom Inventory

Table 1: Full axiom inventory (60 total).

Category	Count	Used by
Type declarations (Model, numTrees, etc.)	6	Infrastructure (Setup.lean)
Core properties (surjective, splitCount, etc.)	6	GBDT bounds
Measure infrastructure	2	Variance (Mathlib connection)
Query complexity constants	2	Query complexity scaling
Universal instances (per-instance configs)	44	Rashomon witnesses

Axiom stratification (verified by `#print axioms`):

- `explanation_impossibility`: **zero** axioms (Rashomon is a hypothesis).
- `attribution_impossibility`: zero behavioral axioms (only Model + attribution types).
- `impossibility_qualitative`: zero behavioral axioms (dominance + surjectivity as hypotheses).
- `gbdt_impossibility_local`: 4 axioms (surjectivity, first-mover, non-first-mover—no proportionality).
- `impossibility` (quantitative): 5 axioms (+ proportionality_global for ratio).
- `consensus_equity` (DASH): 6 axioms (+ cross-group symmetric).
- `attribution_impossibility_bundled`: zero axioms (fully parametric via GBDT-Setup).

Formerly axiomatized, now derived: `spearman_classical_bound`, `le_cam_lower_bound`, `consensus_variance_bound`, `attribution_sum_symmetric`, `attribution_variance`, `attribution_variance_nonneg`, `attribution_proportional`.

9 Defenses Against Common Objections

9.1 "Isn't This Obvious?"

A natural reaction is that practitioners already know explanations are unstable. The contribution is fourfold. (1) We make explicit that six domains share *one* mechanism—the Rashomon property—rather than six unrelated pathologies. (2) The Lean 4 formalization (67 files, 319 theorems, 0 `sorry`) eliminates logical gaps. (3) Empirical experiments quantify severity: 60% attention flip rates, 23.5% counterfactual flip rates, 0.90 concept instability, 80% model selection flip rates. (4) The G-invariant resolution is constructive and actionable.

Arrow's impossibility theorem [Arrow, 1951] was arguably obvious once stated—every committee chair knows voting rules are imperfect. Yet its formalization created the field of social choice theory, precisely because it replaced intuition with a theorem that could be built upon.

9.2 "All Instances Are Trivial"

A reviewer might object that each individual instance is already known. The contribution is not any single instance but the *meta-theorem*: the same four-line proof closes all six cases under a single structural condition. Before this work, a practitioner encountering attention instability might switch to SHAP, then to counterfactual explanations, then to concept probes—each time hoping the new method escapes the problem. Theorem 1 shows no such escape exists: the impossibility is structural (it follows from the Rashomon property), not method-specific. Recognizing this unifies the practitioner's response: rather than debugging each method individually, one diagnoses the Rashomon property and applies the G-invariant resolution uniformly.

References

Kenneth J Arrow. *Social Choice and Individual Values*. Wiley, 1951.

Blair Bilodeau, Natasha Jaques, Pang Wei Koh, and Been Kim. Impossibility theorems for feature attribution. *Proceedings of the National Academy of Sciences*, 121(2):e2304406120, 2024.

Tolga Bolukbasi, Kai-Wei Chang, James Zou, Venkatesh Saligrama, and Adam Kalai. Man is to computer programmer as woman is to homemaker? Debiasing word embeddings. In *Advances in Neural Information Processing Systems*, volume 29, 2016.

Alexandra Chouldechova. Fair prediction with disparate impact: A study of bias in recidivism prediction instruments. *Big Data*, 5(2):153–163, 2017.

Alexander D'Amour, Katherine Heller, Dan Moldovan, et al. Underspecification presents challenges for credibility in modern machine learning. *Journal of Machine Learning Research*, 23(226):1–61, 2022.

Leonardo de Moura and Sebastian Ullrich. The Lean 4 theorem prover and programming language. In *International Conference on Automated Deduction (CADE)*, 2021.

Thomas Decker, Luisa-Marie Herm, Robert Feldhans, Michael Kamp, and Barbara Hammer. Provably better explanations with optimized aggregation of feature attributions. In *International Conference on Machine Learning*, 2024.

Aaron Fisher, Cynthia Rudin, and Francesca Dominici. All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously. *Journal of Machine Learning Research*, 20(177):1–81, 2019.

Andrew Herren and P Richard Hahn. Statistical inference for feature rankings under model multiplicity. *Journal of Computational and Graphical Statistics*, 2023.

Xuanxiang Huang and Joao Marques-Silva. On the failings of Shapley values for explainability. *International Journal of Approximate Reasoning*, 171:109112, 2024.

Sarthak Jain and Byron C Wallace. Attention is not explanation. In *Proceedings of the 2019 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT)*, pages 3543–3556, 2019.

Been Kim, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viegas, and Rory Sayres. Interpretability beyond classification error: Applying testing with concept activation vectors (TCAV). In *International Conference on Machine Learning*, pages 2668–2677, 2018.

Jon Kleinberg, Sendhil Mullainathan, and Manish Raghavan. Inherent trade-offs in the fair determination of risk scores. In *Innovations in Theoretical Computer Science (ITCS)*, 2017.

Gabriel Laberge, Yann Pequignot, Alexandre Mathieu, Foutse Khomh, and Mario Marchand. Partial order in chaos: Consensus on feature attributions in the rashomon set. *Journal of Machine Learning Research*, 24(364):1–50, 2023.

Charles Marx, Flavio Calmon, and Berk Ustun. But are you sure? an uncertainty-aware perspective on explainability. In *Advances in Neural Information Processing Systems*, 2024.

Tobias Nipkow. Social choice theory in HOL: Arrow and Gibbard-Satterthwaite. *Journal of Automated Reasoning*, 43:289–304, 2009.

Miquel Noguer i Alonso. Mathematical foundations of explainability. *SSRN Working Paper*, 2025.

Shrisha Rao. The limits of AI explainability: An algorithmic information theory approach. *arXiv preprint arXiv:2504.20676*, 2025.

Shauli Ravfogel, Yanai Elazar, Hila Gonen, Michael Twiton, and Yoav Goldberg. Null it out: Guarding protected attributes by iterative nullspace projection. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics (ACL)*, pages 7237–7256, 2020.

Cynthia Rudin et al. Position: Amazing things come from having many good models. In *International Conference on Machine Learning (ICML)*, 2024.

Lesia Semenova, Cynthia Rudin, and Ronald Parr. On the existence of simpler machine learning models. *ACM/IMS Journal of Data Science*, 2022.

Thomas Verma and Judea Pearl. Equivalence and synthesis of causal models. In *Conference on Uncertainty in Artificial Intelligence*, pages 255–270, 1991.

Yuanhe Zhang, Jason D Lee, and Fanghui Liu. Statistical learning theory in Lean 4: Empirical processes from scratch. *arXiv preprint arXiv:2602.02285*, 2026. URL `https://arxiv.org/abs/2602.02285`.